Capstone release review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Capstone failure testing. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of capstone release review. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of capstone **release review**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone failure testing. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Documentation and operational readiness. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Performance and cost experiments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of capstone **release review**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Capstone failure testing. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain capstone **release review** using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Capstone release review case. Separate observations, given facts and assumptions before choosing an explanation.

Release review combines correctness, access, failure recovery and operational readiness. **Critical gate**s can block release independently of an average score.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

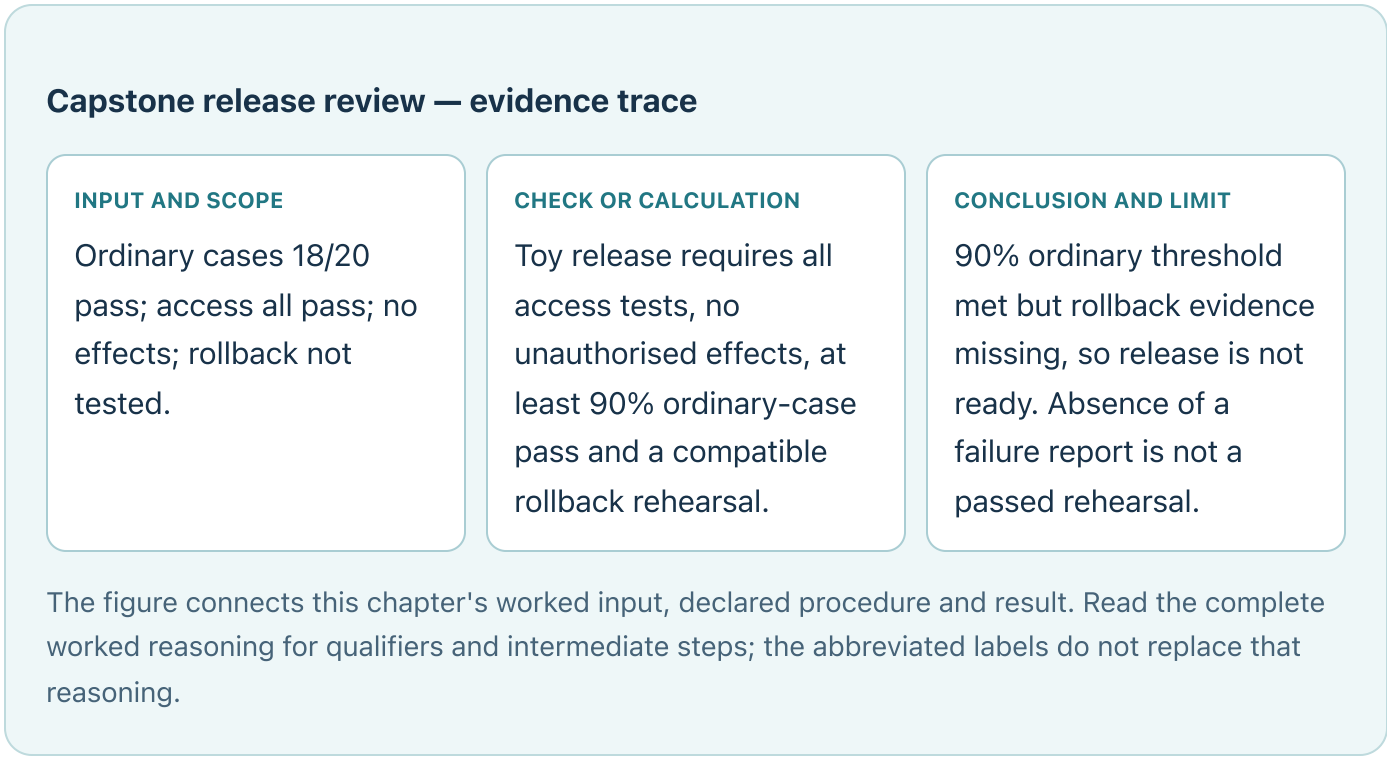
These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Capstone release review. Connect each labelled input, deciding step and result.

Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible **rollback rehearsal**. Record each gate separately with versioned **evidence**.



Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Capstone release review. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Every listed deletion location passes new verification.

Predicted result

Prediction: That gate can pass; review the complete current **evidence** before approving release.

Why the result follows

Reason: One repaired gate does not automatically refresh **evidence** for every other gate.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Capstone release review to explain a concrete decision, rather than

only memorising definitions.

Word	Meaning
Release review	A decision using versioned gate evidence
Critical gate	A requirement that cannot be averaged away
Rollback rehearsal	A tested recovery of a compatible version set

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Capstone release review, predict a step, then compare with its explanation.

Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested.

90% ordinary threshold met but rollback **evidence** missing, so release is not ready. Absence of a failure report is not a passed rehearsal.

Worked explanation · Capstone release review

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone release review

Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass.

The facts we will use

> **Supplied facts**
>
> Factual and access checks pass. Deletion fails because a listed backup remains. All three gates are mandatory.

AI ACADEMY / AI-6/35

Capstone release review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. List evidence

Record two passes and the specific deletion failure.



2. Apply decision rule

Block release; a majority of passes is insufficient.



3. Specify repair

Remove the listed copy and verify it before a fresh review.

CHANGE ONE THING

Every listed deletion location passes new verification.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. List evidence	Record two passes and the specific deletion failure.
2. Apply decision rule	Block release; a majority of passes is insufficient.
3. Specify repair	Remove the listed copy and verify it before a fresh review.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Release review applies explicit conditions to a specific candidate version. Completed coding is not **evidence** that all operational obligations are satisfied.

What this example does not establish

Keep the stated limit

Approval cannot guarantee zero future faults; retain ownership, monitoring and recovery arrangements.

Keep the stated limit

Read the labelled picture: Apply the three mandatory release gates to the recorded results. Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?

Keep the stated limit

Change one thing: Every listed deletion location passes new verification. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Capstone release review claim and explain a repair using the supplied evidence.

Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Capstone release review with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone **release review**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone **release review**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Apply the three mandatory release gates to the recorded results. Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Capstone release review. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Capstone release review. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

Study a complete case

> Revisit the complete Capstone release review case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Capstone release review — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested.	Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible rollback rehearsal.	90% ordinary threshold met but rollback evidence missing, so release is not ready. Absence of a failure report is not a passed rehearsal.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Capstone release review, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass.

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone release review	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Capstone release review without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone **release review**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Remove the listed copy and verify it before a fresh review.

1. List **evidence**

2. Specify repair

3. Apply decision rule

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Specify repair Remove the listed copy and verify it before a fresh review. **Release review** applies explicit conditions to a specific candidate version.

Completed coding is not **evidence** that all operational obligations are satisfied.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Every listed deletion location passes new verification. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. That gate can pass; review the complete current **evidence** before approving release.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That gate can pass; review the complete current **evidence** before approving release. One repaired gate does not automatically refresh **evidence** for every other gate.

Repair a missing building block

Review Capstone failure testing (ai-6/34). Goal: Explain the **mechanism** and terminology of capstone failure testing. Short practice: Explain why retrieval success permits the checked 15:00 fallback when drafting fails.

What changes if retrieval fails too, and why can a template not fill that **evidence** gap?. Return to this week and explain what you changed.

Review Documentation and operational readiness (ai-6/32). Goal: Explain the **mechanism** and terminology of documentation and operational readiness. Short practice: Match SOURCE_UNAVAILABLE to its safe check and escalation owner.

What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?. Return to this week and explain what you changed.

Review Performance and cost experiments (ai-6/29). Goal: Explain the **mechanism** and terminology of performance and cost experiments. Short practice: Calculate per-case seconds and cost units for A and B.

Which meets a budget of at most 2 cost units per case, and what speed trade-off does that choice make?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

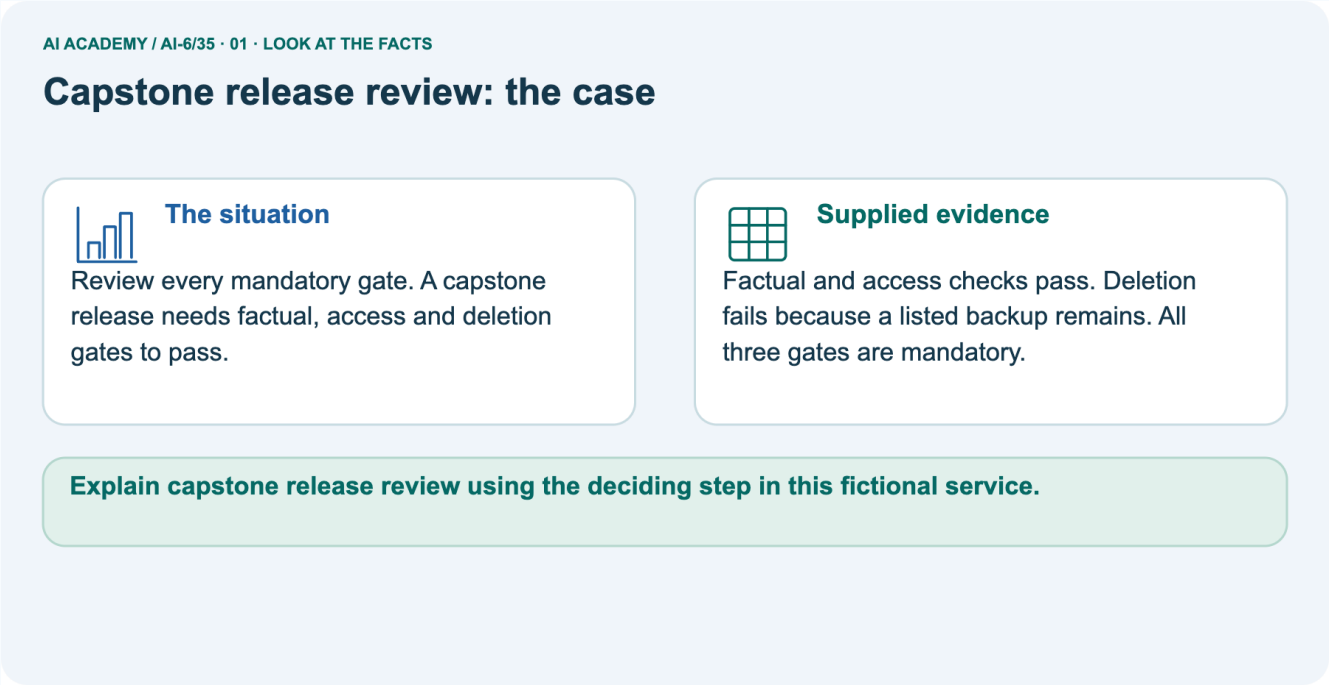
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Capstone release review graphic using its labels, deciding evidence and relevant limitation.

01. Capstone release review: the case

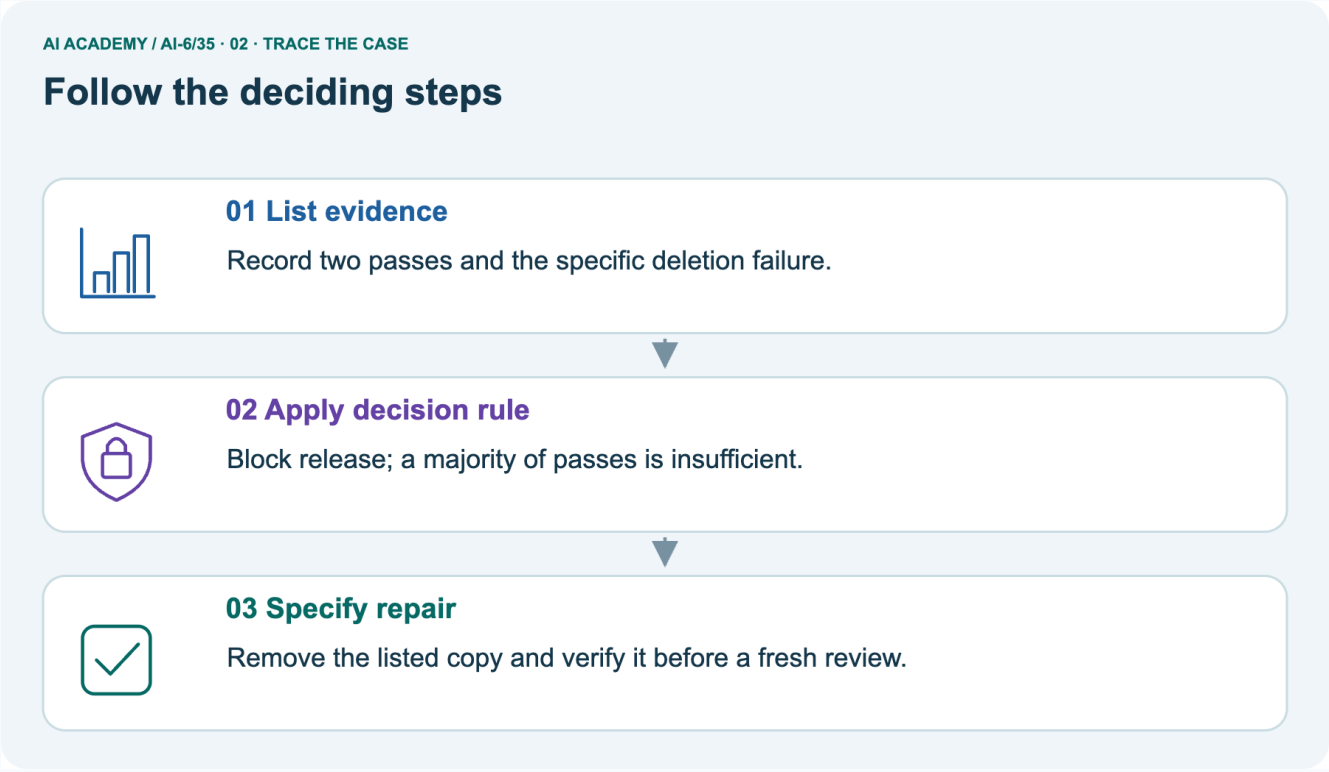


Follow the labelled picture with the matching explanation. Open the vector diagram

Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass. Supplied facts: Factual and access checks pass. Deletion fails because a listed backup remains. All three gates are mandatory.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

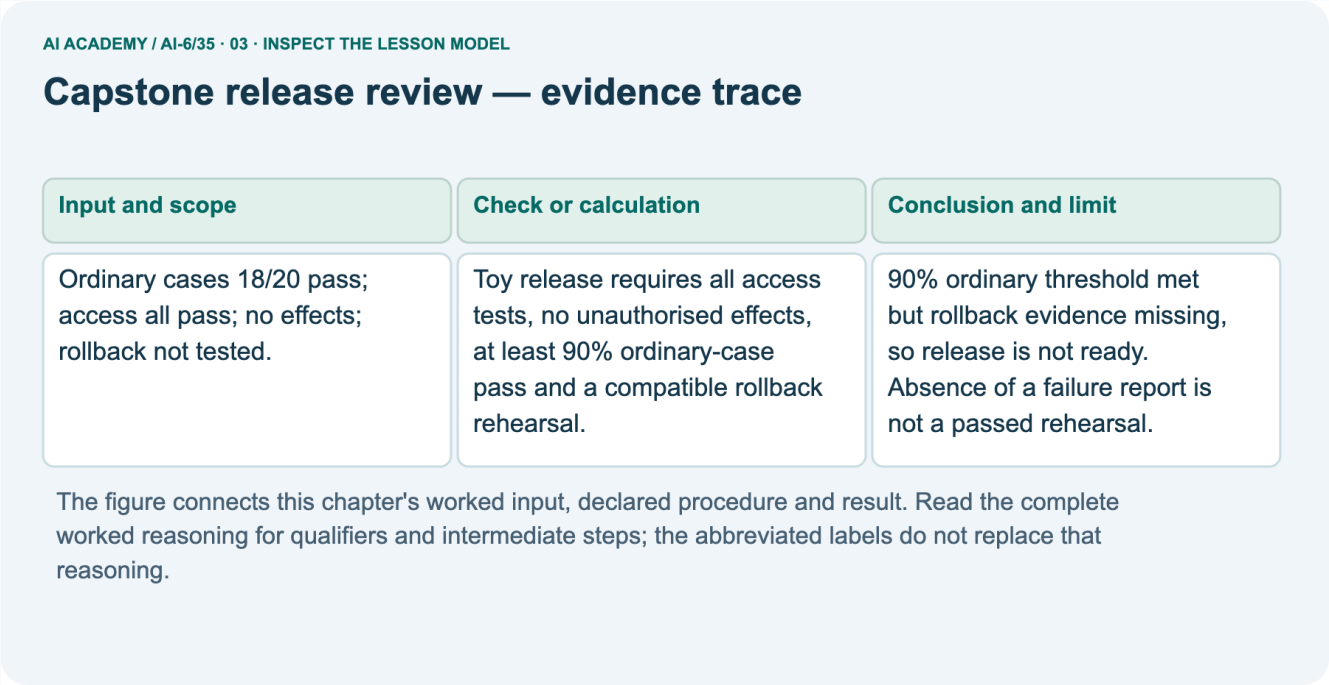


Follow the labelled picture with the matching explanation. Open the vector diagram

List **evidence**: Record two passes and the specific deletion failure. → Apply decision rule: Block release; a majority of passes is insufficient. → Specify repair: Remove the listed copy and verify it before a fresh review.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone release review — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

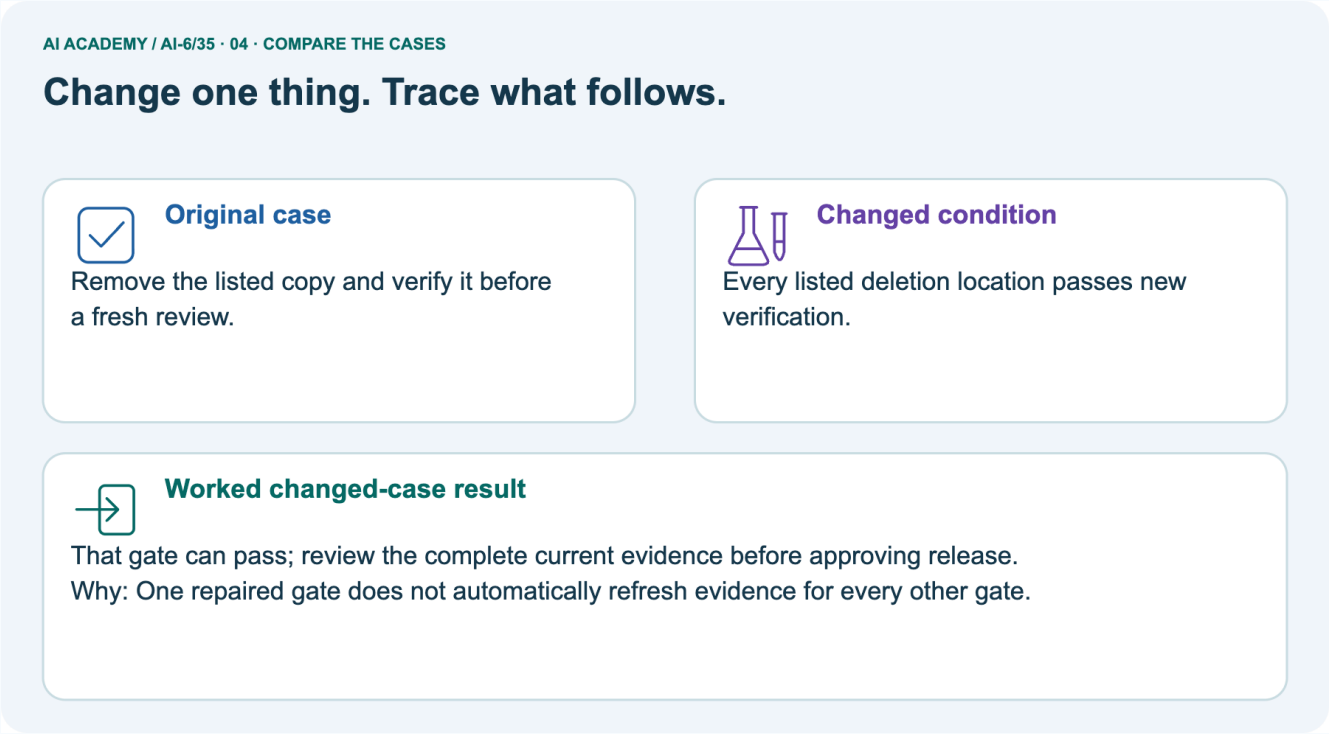
Capstone **release review** — **evidence** trace.

Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. | Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible **rollback rehearsal**. | 90% ordinary threshold met but rollback **evidence** missing, so release is not ready.

Absence of a failure report is not a passed rehearsal.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



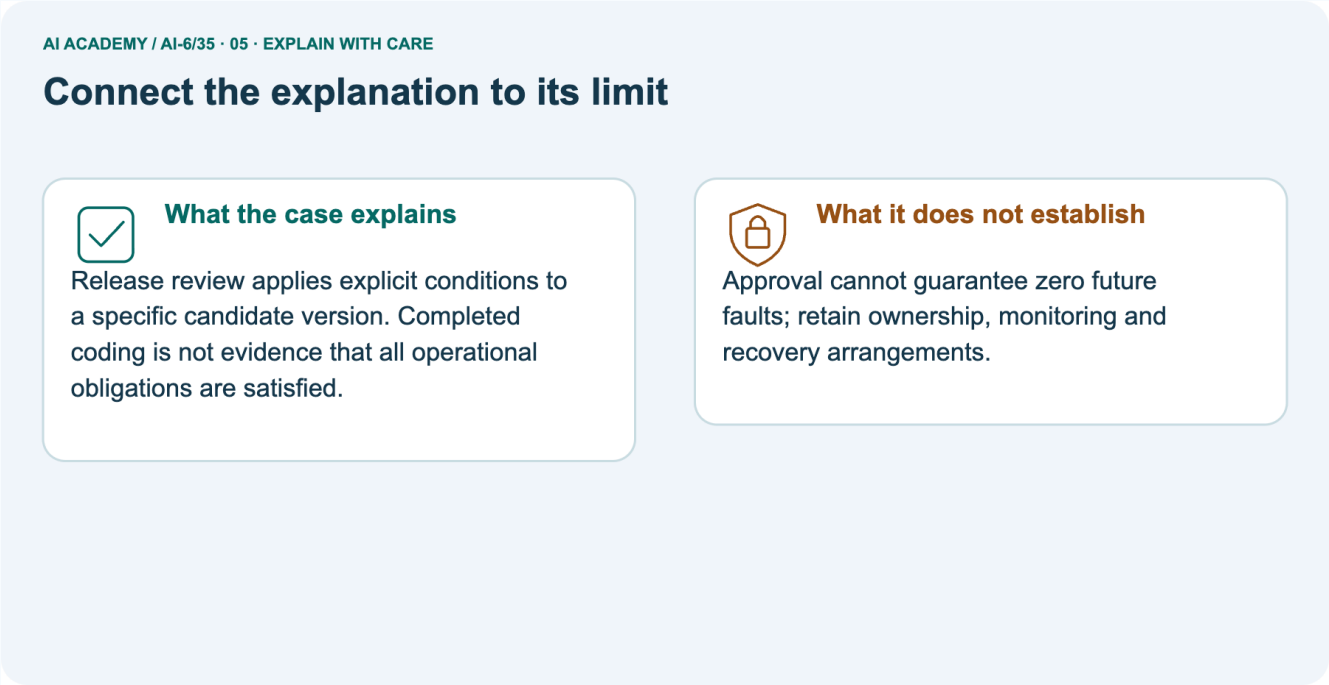
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Remove the listed copy and verify it before a fresh review. Changed condition: Every listed deletion location passes new verification.

Expected result: That gate can pass; review the complete current **evidence** before approving release. Reason: One repaired gate does not automatically refresh **evidence** for every other gate.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



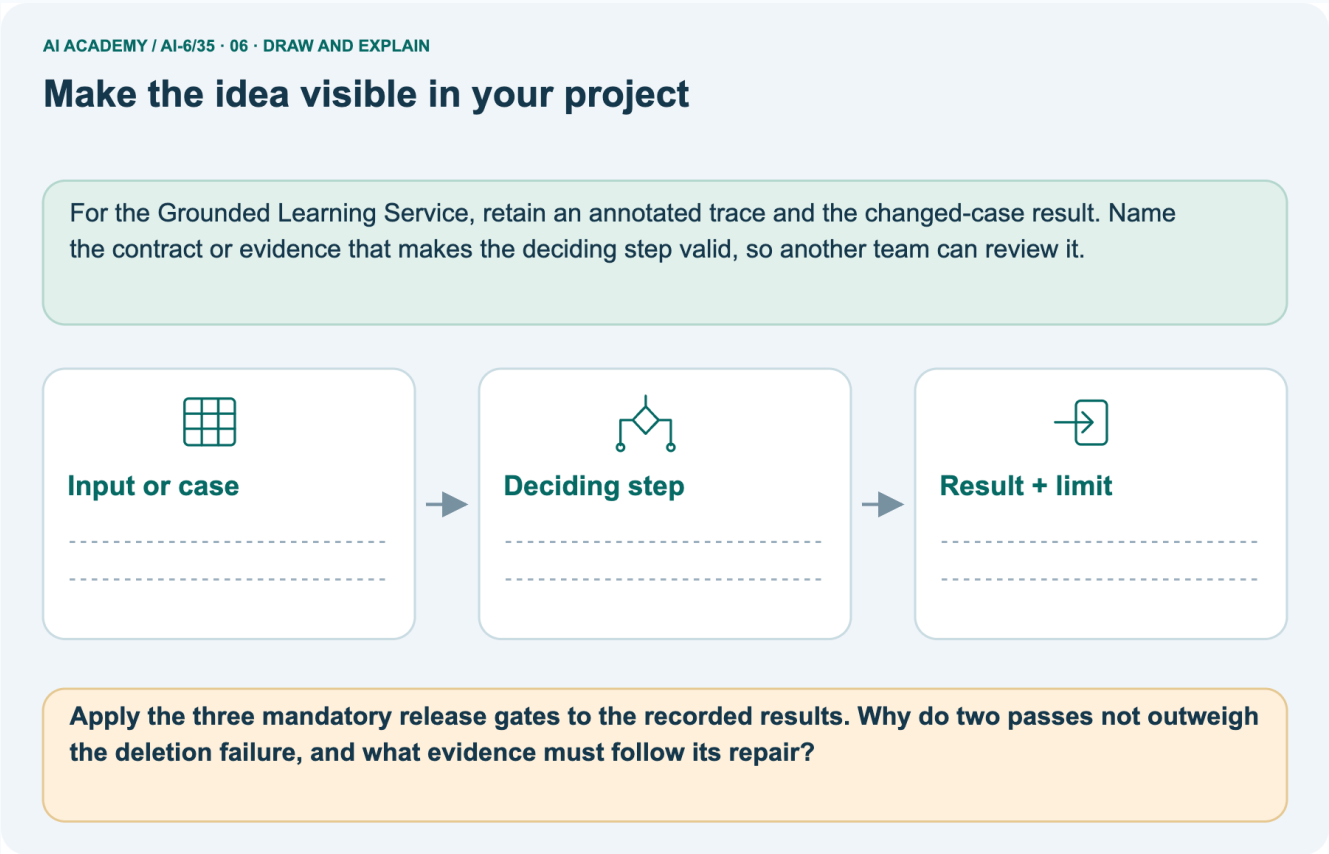
Follow the labelled picture with the matching explanation. Open the vector diagram

Release review applies explicit conditions to a specific candidate version. Completed coding is not **evidence** that all operational obligations are satisfied.

Limit: Approval cannot guarantee zero future faults; retain ownership, monitoring and recovery arrangements.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the three mandatory release gates to the recorded results.

Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.